Week 10 MAT 110

PART 2 INTRO TO POLAR COORDIATES (CYLINDRICAL, SPHERICAL) [EXAMPLES]

Examples

1. Let $(r, \theta, z) = (4, \frac{\pi}{3}, -3)$ → cylindrical coordinate.

Evaluate the rectangular coordinate (x, y, z).

$x = r\cos\theta = 4\cos\frac{\pi}{3} = 4(\frac{1}{2}) = 2$

$y = r\sin\theta = 4\sin\frac{\pi}{3} = 4(\frac{\sqrt{3}}{2}) = 2\sqrt{3}$

$z = -3$

$\therefore (x, y, z) = (2, 2\sqrt{3}, -3)$

2. Given that $(\rho, \theta, \varphi) = (4, \frac{\pi}{3}, \frac{\pi}{4})$ → spherical coordinate.

Evaluate (x, y, z), the rectangular coordinate

$x = \rho\sin\varphi\cos\theta = 4\sin\frac{\pi}{4}\cos\frac{\pi}{3} = 4 \times \frac{1}{\sqrt{2}} \times \frac{1}{2} = \sqrt{2}$

$y = \rho\sin\varphi\sin\theta = 4\sin\frac{\pi}{4}\sin\frac{\pi}{3} = 4(\frac{1}{\sqrt{2}})(\frac{\sqrt{3}}{2}) = \sqrt{2}\sqrt{3} = \sqrt{6}$

$z = \rho\cos\varphi = 4\cos\frac{\pi}{4} = 4(\frac{1}{\sqrt{2}}) = 2\sqrt{2}$

$(x, y, z) = (\sqrt{2}, \sqrt{6}, 2\sqrt{2})$

Examples

1 Convert $\left(-4, \frac{2\pi}{3}\right)$ into Cartesian coordinates.

Given $r = -4$, $\theta = \frac{2\pi}{3}$

$$x = r\cos\theta = -4\cos\frac{2\pi}{3} = (-4)\left(-\frac{1}{2}\right) = +2$$

$$y = r\sin\theta = -4\sin\frac{2\pi}{3} = (-4)\left(\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$$

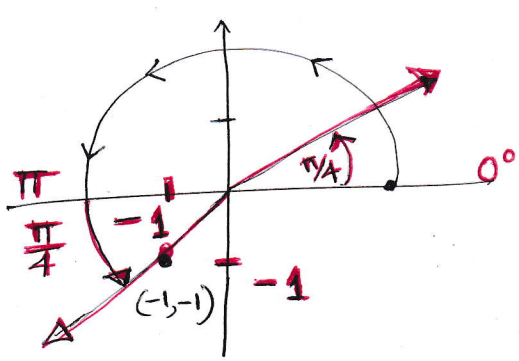
$$(x, y) = (+2, -2\sqrt{3})$$

2 Convert $(-1, -1)$ into Polar coordinates.

$x = -1$, $y = -1$

$$r = \sqrt{x^2+y^2} = \sqrt{(-1)^2+(-1)^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\frac{y}{x} = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

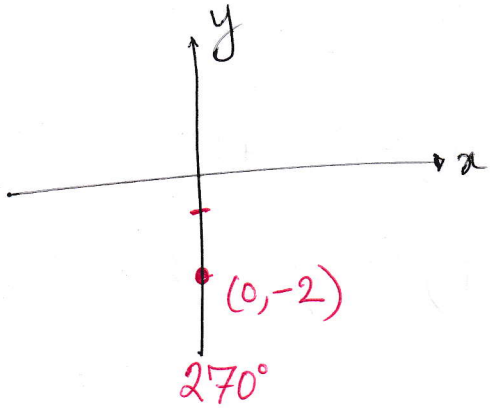


$$\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\therefore (r, \theta) = \left(\sqrt{2}, \frac{5\pi}{4}\right)$$

$\boxed{3}$ Convert into polar coordinates: $(0,-2)$.

$x = 0, \quad y = -2$



$$r = \sqrt{x^2+y^2}$$
$$= \sqrt{0^2+(-2)^2} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$
$$= \tan^{-1}\left(\frac{-2}{0}\right)$$
$$= \tan^{-1}(\infty)$$
$$= 90^\circ \text{ or } 270^\circ$$
$$\theta = 270^\circ \text{ or } \frac{3\pi}{2}$$

$$(r, \theta) = \left(2, \frac{3\pi}{2}\right)$$

Find the spherical coordinates of the point $(4, -4, 4\sqrt{6})$ which is in rectangular coordinates.

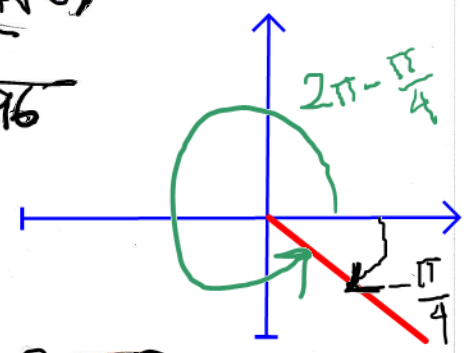
Given: $(x, y, z) = (4, -4, 4\sqrt{6})$

$$\rho = \sqrt{x^2+y^2+z^2} = \sqrt{4^2+(-4)^2+(4\sqrt{6})^2}$$

$$= \sqrt{16+16+96}$$

$$= \sqrt{128}$$

$$= \sqrt{64 \times 2} = 8\sqrt{2}$$



$\theta = \theta_0 \pm 2n\pi$

$$\theta = \tan^{-1}\frac{y}{x} = \tan^{-1}\left(\frac{-4}{4}\right) = \tan^{-1}(-1) = -\frac{\pi}{4}$$

But $\theta \in (0, 2\pi)$ $\therefore \theta = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$

$$\cos\varphi = \frac{|z|}{\rho} = \frac{|4\sqrt{6}|}{\sqrt{128}} = \frac{4\sqrt{6}}{\sqrt{128}}$$

$$\varphi = \cos^{-1}\left(\frac{4\sqrt{6}}{\sqrt{128}}\right)$$

$$= 30^\circ \text{ or } \frac{\pi}{6}$$

$$\therefore (\rho, \theta, \varphi) = \left(8\sqrt{2}, \frac{7\pi}{4}, \frac{\pi}{6}\right)$$